

DG Interns Hub

Cybersecurity Internship – Week 2

Advanced Networking, Analysis & Security Concepts

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Internship: Cybersecurity Internship – Week 2

Focus: Advanced Networking, Analysis & Security Concepts

Practical Tools: Wireshark, Nmap, Cisco Packet Tracer

Week 2 Practical & Theory Report

1. Introduction & Objectives

This report documents the Week 2 cybersecurity internship activities focused on advanced networking, network traffic analysis, authorized scanning, LAN configuration and safe understanding of attack concepts.

Objectives

- Understand common network ports, protocols and services.
- Understand the roles of routers, switches and firewalls.
- Analyze DNS and encrypted TLS traffic using Wireshark.
- Perform basic authorized service discovery using Nmap.
- Build and test a small LAN in Cisco Packet Tracer.
- Understand MITM, DoS and packet sniffing concepts and their prevention.

Tools Used

Wireshark – packet capture and protocol analysis.

Nmap – authorized network/service discovery.

Cisco Packet Tracer – network topology and connectivity simulation.

2. Ports & Services

Ports are logical communication endpoints. Services commonly listen on well-known port numbers so that network traffic can be delivered to the appropriate application.

Port	Protocol	Service	Usage
20	TCP	FTP Data	File transfer data
21	TCP	FTP	File transfer/control
22	TCP	SSH	Secure remote login
23	TCP	Telnet	Remote login
25	TCP	SMTP	Sending email
53	TCP/UDP	DNS	Domain name resolution
67	UDP	DHCP Server	IP configuration
68	UDP	DHCP Client	Receives IP configuration
80	TCP	HTTP	Web traffic
110	TCP	POP3	Email retrieval
123	UDP	NTP	Time synchronization
143	TCP	IMAP	Email retrieval
161	UDP	SNMP	Network management
389	TCP/UDP	LDAP	Directory services
443	TCP	HTTPS	Secure web traffic
445	TCP	SMB	File/printer sharing
3389	TCP	RDP	Remote Desktop
3306	TCP	MySQL	Database service
5432	TCP	PostgreSQL	Database service
8080	TCP	HTTP-alt	Alternative web service

3. Networking Devices

Router

A router connects different networks and forwards packets according to destination IP addresses and routing information. A typical home router connects a private LAN to the Internet.

Switch

A switch connects devices within a LAN and forwards Ethernet frames using MAC-address information. It is commonly used to connect PCs, printers and servers in a local network.

Firewall

A firewall controls network traffic according to security rules. It can allow legitimate traffic and block unauthorized or suspicious connections.

Router vs Switch

Router: connects networks and makes forwarding decisions using IP information. Switch: connects devices in a LAN and primarily forwards frames using MAC information.

4. Network Security Basics

Network Security

Network security protects networks, systems and data against unauthorized access, misuse, disruption and attacks. The main security goals are confidentiality, integrity and availability.

Man-in-the-Middle (MITM)

Conceptually, an attacker attempts to position themselves between two communicating parties so traffic can be observed or manipulated. Public Wi-Fi is a common example of an environment where users may face increased interception risks.

DoS (Denial of Service)

A DoS attack attempts to make a service unavailable by overwhelming or exhausting resources. Defenses include filtering, rate limiting, monitoring and DDoS protection.

Packet Sniffing

Packet sniffing means capturing and analyzing network packets. It has legitimate uses in troubleshooting and security analysis, but unencrypted traffic can expose sensitive information.

Basic Prevention

Use HTTPS/TLS, VPNs on untrusted networks, secure Wi-Fi, strong authentication and network monitoring. Do not ignore certificate/security warnings.

5. Practical – Wireshark Packet Analysis

The Wireshark capture shows real traffic generated by the system. The screenshots identify DNS requests/responses and TLS/HTTPS-related traffic.

DNS Analysis

The DNS-filtered capture shows the local host 192.168.1.7 communicating with DNS server 192.168.1.1. Examples include lookups for web.whatsapp.com and chatgpt.com.

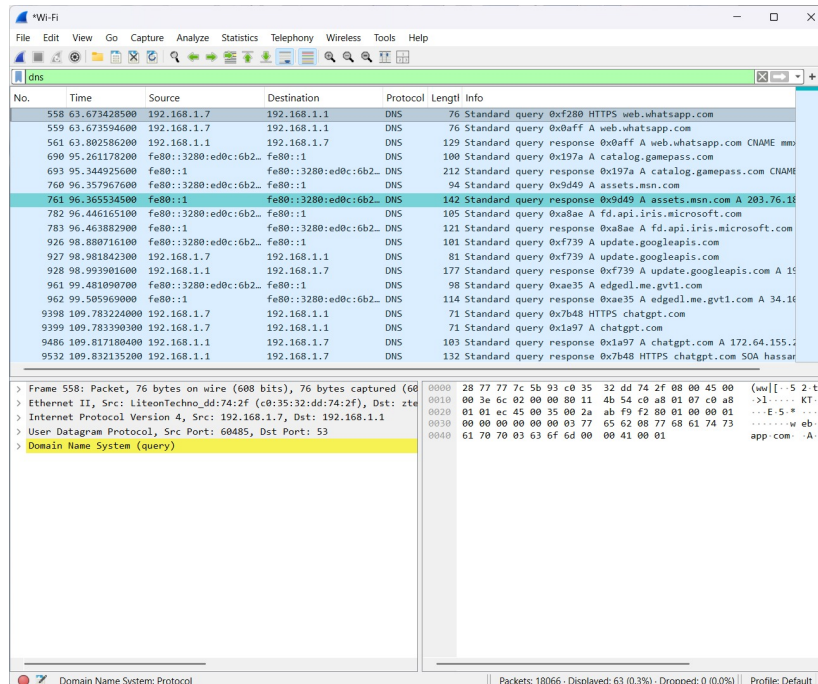


Figure 1: Wireshark DNS-filtered packet capture.

5.1 Wireshark – TLS/HTTPS Traffic

The TLS filter displays encrypted application traffic. One selected packet shows source 192.168.1.7, destination 130.211.32.14 and destination TCP port 443.

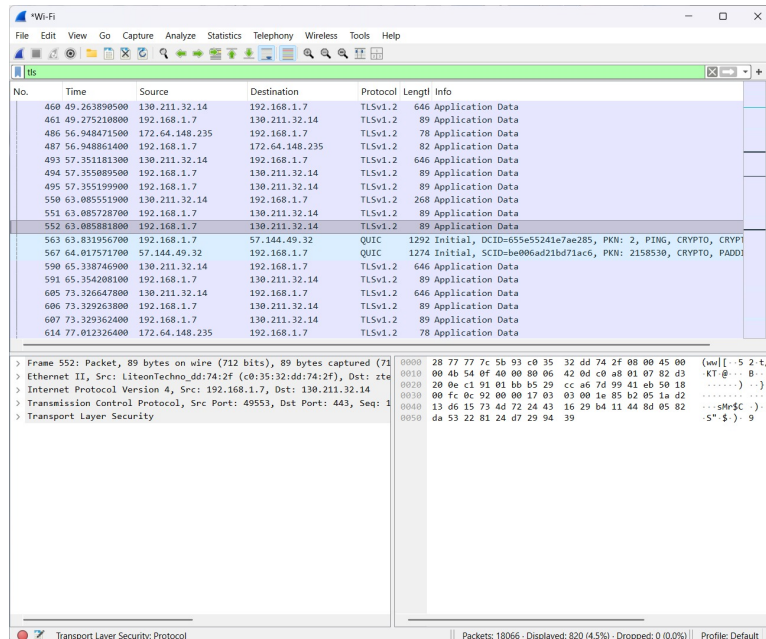


Figure 2: TLS-filtered traffic in Wireshark.

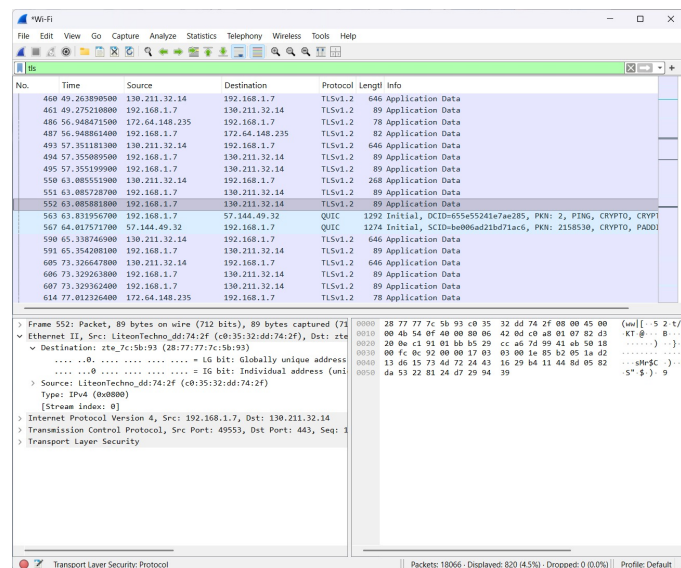


Figure 3: Selected TLS packet details showing source/destination and port 443.

6. Practical – Nmap Network Scanning

Nmap was used against the authorized practice target scanme.nmap.org. The first scan identifies open TCP ports and the second uses -sV for service/version detection.

Command 1: nmap scanme.nmap.org

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\AMAR PRATAP> nmap scanme.nmap.org
Starting Nmap 7.991 ( https://nmap.org ) at 2026-09-09 10:23 +0530
Warning: Hostname scanme.nmap.org resolves to 2 IPs. Using 45.33.32.156.
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.27s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
9929/tcp  open  nping-echo
31337/tcp open  Elite

Nmap done: 1 IP address (1 host up) scanned in 5.46 seconds
PS C:\Users\AMAR PRATAP> |
```

Figure 4: Basic Nmap scan showing open ports.

Command 2: nmap -sV scanme.nmap.org

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\AMAR PRATAP> nmap -sV scanme.nmap.org
Starting Nmap 7.991 ( https://nmap.org ) at 2026-09-09 10:25 +0530
Warning: Hostname scanme.nmap.org resolves to 2 IPs. Using 45.33.32.156.
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.27s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
rDNS record for 45.33.32.156: 156.32.33.45.in-addr.arpa
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http         Apache httpd 2.4.7 ((Ubuntu))
9929/tcp  open  nping-echo   Nping echo
31337/tcp open  tcpwrapped

Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.71 seconds
PS C:\Users\AMAR PRATAP> |
```

Figure 5: Nmap service/version detection.

7. Practical – Cisco Packet Tracer LAN

A small network topology was created in Cisco Packet Tracer. The screenshots show PCs connected through a switch, with a router also visible in the broader topology.

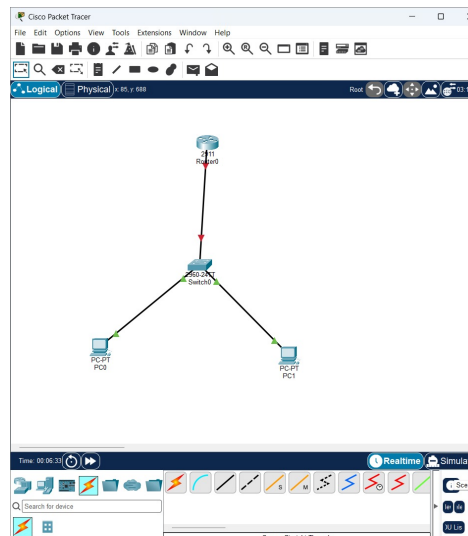


Figure 6: Packet Tracer topology.

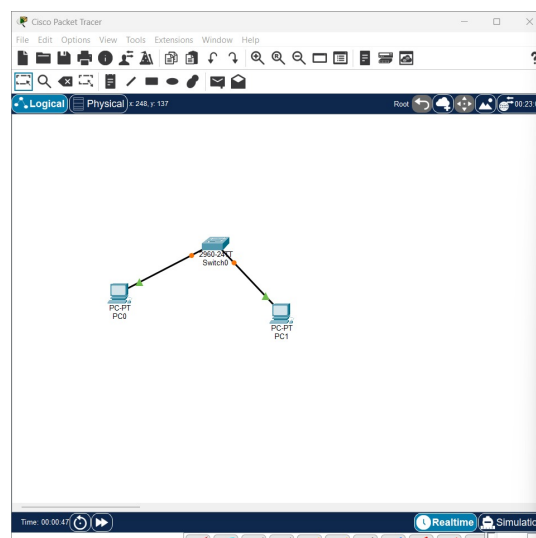
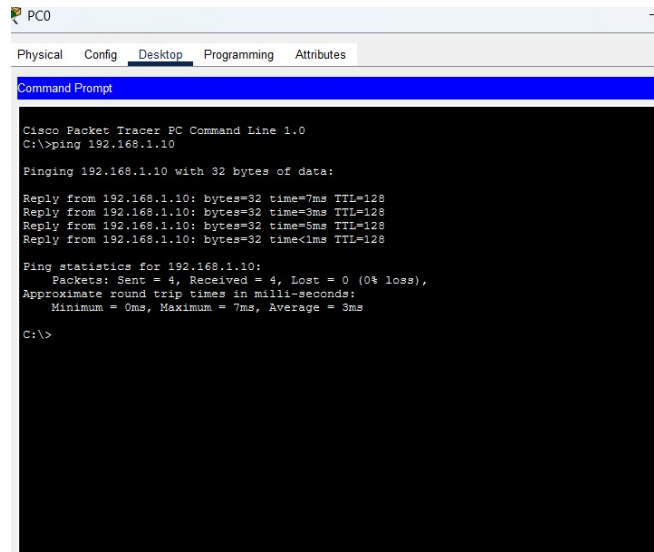


Figure 7: Two PCs connected through a Cisco 2960 switch.

7.1 LAN Connectivity Test

The supplied connectivity screenshots show successful ICMP ping responses with 0% packet loss; however, each shown command pings the same PC's own IP, so they do not prove PC-to-PC connectivity.



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt

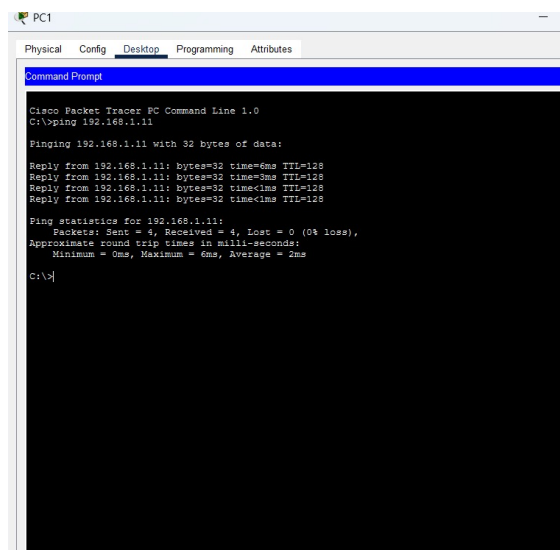
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:
Reply from 192.168.1.10: bytes=32 time=7ms TTL=128
Reply from 192.168.1.10: bytes=32 time=3ms TTL=128
Reply from 192.168.1.10: bytes=32 time=5ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 3ms

C:\>
```

Figure 8: PC0 ping screenshot.



```
PC1
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:
Reply from 192.168.1.11: bytes=32 time=6ms TTL=128
Reply from 192.168.1.11: bytes=32 time=3ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 2ms

C:\>
```

Figure 9: PC1 ping screenshot.

8. MITM Safe Conceptual Simulation

A safe MITM exercise focuses on understanding the communication path rather than capturing real credentials or attacking real users. Conceptually, an attacker attempts to become an intermediary between a victim and the intended server/network.

Conceptual topology: PC1 (Victim) ↔ Switch ↔ PC2 (Attacker) ↔ Router/Server

Risks: Data exposure, session information theft, credential exposure and traffic manipulation.

Prevention: HTTPS/TLS, VPN on untrusted networks, secure Wi-Fi, MFA and careful certificate validation.

9. Learning, Challenges & Conclusion

Learning: I learned how ports map to network services, how routers and switches differ, how Wireshark can reveal DNS/TLS traffic, how Nmap can identify exposed services on an authorized target, and how a basic LAN can be configured and tested.

Challenges: Understanding packet fields and interpreting scan output required careful observation.

How I solved them: I used protocol filters in Wireshark, compared Nmap outputs, and verified connectivity using ping in Packet Tracer.

Conclusion: The Week 2 activities strengthened practical networking and cybersecurity fundamentals while keeping testing within safe and authorized environments.